

SpaceX Falcon 9 First-Stage Landing Prediction

Data Science Capstone Project Report

GitHub Repository: <https://github.com/razerblade09/spacex-falcon9-capstone>

Contains all completed notebooks, Python scripts (data collection, web scraping, wrangling, EDA, SQL, Folium, ML modeling) and the Plotly Dash dashboard app referenced in this presentation.

Executive Summary

Project links — GitHub repository: <https://github.com/razerblade09/spacex-falcon9-capstone> (all notebooks, scripts, and dashboard code)

Methods used: Data was collected via the SpaceX REST API and Wikipedia web scraping, cleaned and labeled, explored with visualization and SQL, mapped geospatially with Folium, presented interactively through a Plotly Dash dashboard, and modeled with four classifiers (Logistic Regression, SVM, Decision Tree, KNN) tuned via 10-fold GridSearchCV.

Key results: Across 90 Falcon 9 launches (2010–Nov 2020) the overall first-stage landing success rate was 66.7%. Logistic Regression, SVM and KNN each reached 83.3% test accuracy; the Decision Tree reached 77.8%. Landing success rises with booster maturity and falls with orbit energy.

90

Launches analyzed

66.7%

Overall landing success

83.3%

Best model test accuracy

4

ML models compared

This presentation completes every required section: Executive Summary, Introduction, Data Collection (SpaceX API and web scraping), Data Wrangling Methodology, EDA with Data Visualization, EDA with SQL queries and results, Folium Map, Plotly Dash dashboard, Predictive Analysis with classification results and confusion matrix, and Conclusion with insights — each on its own slide with charts, tables and analysis.

Conclusion: landing success is highly predictable from launch parameters, so a competing launch provider could reliably estimate SpaceX's true reusable-launch cost advantage.

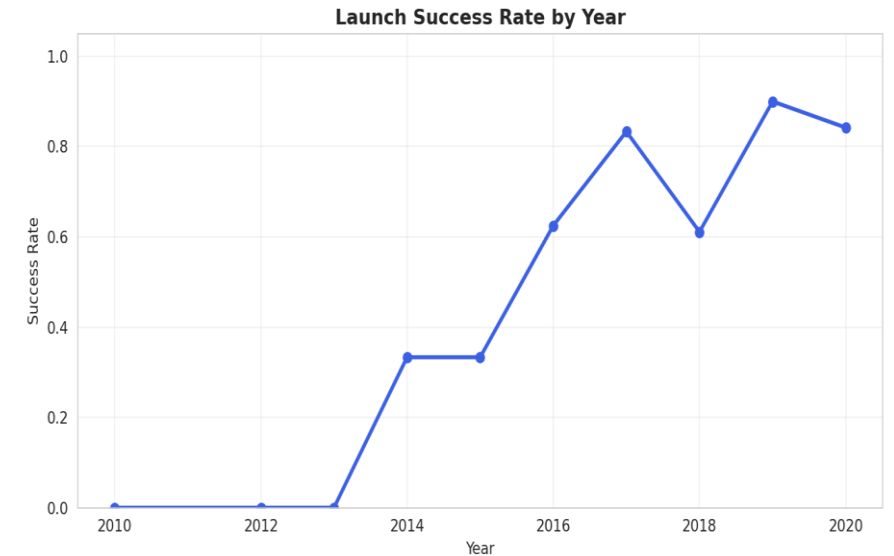
Introduction

Background: SpaceX advertises Falcon 9 launches at \$62 million, versus \$165 million or more from other providers — largely because SpaceX recovers and reuses the rocket's first stage instead of discarding it after every flight.

Problem statement: If the first stage does not land successfully, that launch's true cost changes substantially. Acting as a data scientist for a startup competing with SpaceX, this project predicts from public launch parameters whether the first stage will land, so a competing bid can be costed accurately.

Objectives:

- Collect and clean historical Falcon 9 launch data from the SpaceX API and Wikipedia
- Identify which launch parameters correlate with landing success
- Visualize patterns across launch sites, orbits, payload mass and time
- Build, tune and evaluate a landing-outcome classifier



Falcon 9 landing success climbed from roughly 0% across 2010–2013 to close to 100% by 2019–2020 as SpaceX matured its recovery process, which is what makes the outcome predictable at all.

Data Collection – SpaceX API

Pipeline:

1

GET /v4/launches/past → all historical launches

2

Filter to single-core, single-payload Falcon 9 flights

3

Call /rockets, /launchpads, /payloads and /cores per launch

4

Assemble one row per launch: booster, orbit, site, outcome, mass

```
api.spacexdata.com/v4/launches/past
```

Result: 90 clean Falcon 9 launch records spanning 2010 to November 2020, one row per launch, each carrying booster version, payload mass, orbit, launch site with coordinates, flight count, grid fins, reused flag, legs, landing pad, booster block and serial, and landing outcome.

Falcon 1 launches were excluded — this project addresses only the Falcon 9 landing-prediction problem.

Data Collection – Web Scraping

Pipeline:

- 1 Request the Wikipedia "List of Falcon 9 and Falcon Heavy launches" page
- 2 Parse HTML tables with BeautifulSoup, locating the witable elements
- 3 Extract column headers, then walk each row of launch data
- 4 Clean dates, booster serials and payload mass into a dataframe

```
BeautifulSoup(response.text, 'html.parser')
```

This gives an independent, cross-checked source for the launch records: it validates the API dataset, fills in fields the API does not expose such as payload customer, and supplies the table that is loaded into SQL for the EDA-with-SQL section.

Key techniques: regex and Unicode cleanup of payload-mass strings, plus helper functions that pull booster version and landing status out of merged table cells.

Data Wrangling Methodology

Missing values: PayloadMass was missing for 5.6% of records and was filled with the column mean. LandingPad was missing for 28.9% and was deliberately kept as-is, because a null pad legitimately means no ground or drone-ship landing was attempted.

Label engineering: The raw Outcome field — free text such as "True ASDS", "False Ocean" or "None None" — was converted into a binary Class label: 1 if the booster landed successfully by any method (drone ship, ground pad, RTLS return or controlled ocean), 0 if it did not land or no attempt was made.

Result: of 90 launches, 60 (66.7%) ended in a successful first-stage landing and 30 (33.3%) did not. This Class column is the target variable for the predictive model.

60

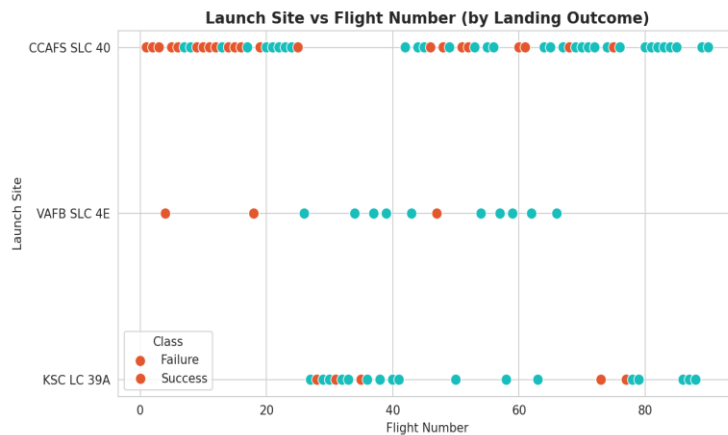
successful landings

30

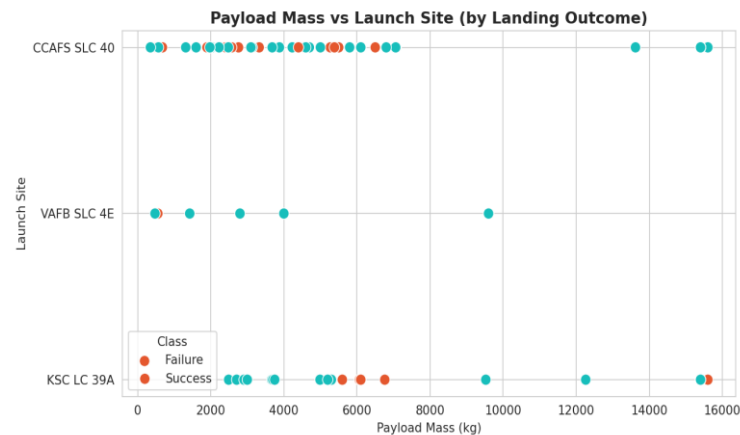
unsuccessful

EDA with Data Visualization

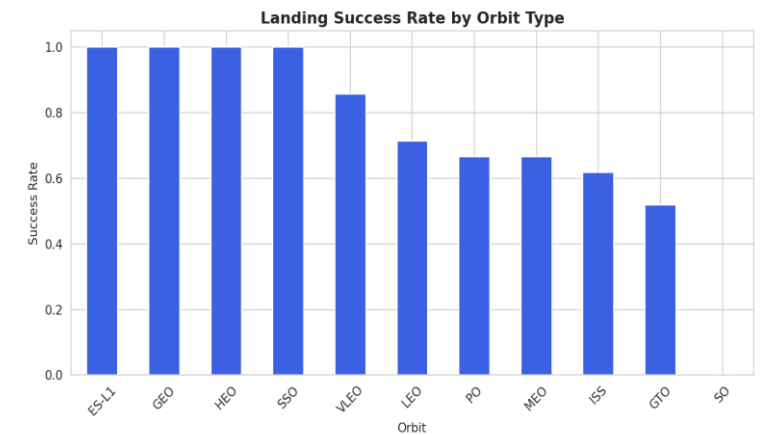
Flight number, payload mass, orbit type and launch year were plotted against landing outcome to surface which factors correlate with success. Six charts were produced: scatter plots of flight number against launch site and against orbit, scatter plots of payload mass against launch site and against orbit, a bar chart of success rate by orbit, and a line chart of success rate by year.



Flight # vs Launch Site



Payload vs Launch Site



Success Rate by Orbit

Findings: heavier, higher-energy GTO and ISS missions land least reliably, while light LEO, SSO and GEO missions land most reliably. Success rate climbs steadily with flight number at every site, and rises from roughly 0% across 2010–2013 to close to 100% by 2019–2020 — landing success is a maturing process, not a fixed property of the hardware.

EDA with SQL

The scraped launch table was loaded into a SQL database (SQLite / IBM Db2) as SPACEXTABLE and queried to answer specific business questions:

- Distinct launch sites used
- Total payload mass carried for NASA (CRS)
- Average payload mass for F9 v1.1 boosters
- Date of the first successful ground-pad landing
- Drone-ship successes carrying 4,000–6,000 kg
- Count of each mission outcome
- Booster(s) carrying the maximum payload mass
- Failed drone-ship landings during 2015
- Ranking of landing outcomes, 2010–2017

Query	Result
Distinct launch sites	CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
Total payload, NASA (CRS)	45,596 kg
Avg. payload, F9 v1.1	2,534.7 kg
First ground-pad landing	1 May 2017
Max payload carried	15,600 kg (F9 B5)
Failed drone-ship, 2015	2 (B1012, B1015)
Top outcome, 2010–2017	Success (20), No attempt (10)

These query results ground the visual EDA in exact, verified numbers: four active launch sites, a clear NASA CRS payload total, and a landing-outcome ranking in which outright successes already outnumber no-attempts by 2017.

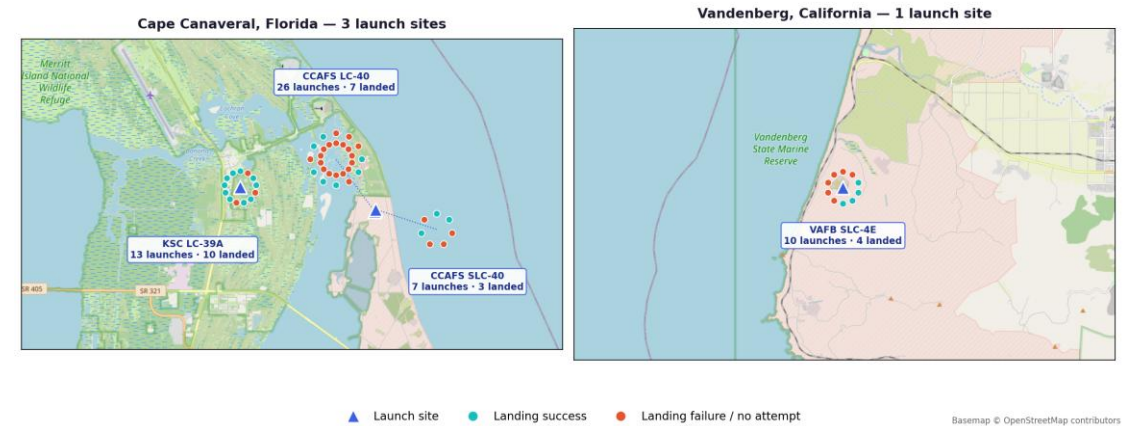
Folium Map

All 4 launch sites are plotted with markers, and every individual launch (56 records) is plotted as a marker colored green for a successful landing and red for a failed or no-attempt landing, so denser green clusters are visible per site.

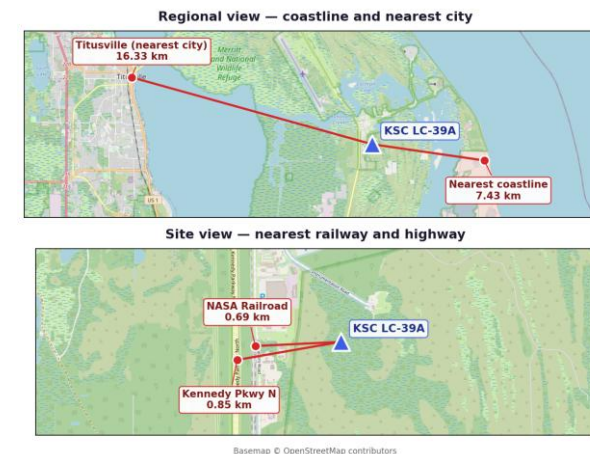
Observation: KSC LC-39A shows the densest green cluster, landing 10 of 13 launches (77%), while CCAFS LC-40 — the earliest and highest-volume pad — landed only 7 of 26 (27%), because most of its records predate a working recovery process.

Proximity analysis (haversine distance from KSC LC-39A): nearest coastline 7.43 km, nearest city Titusville 16.33 km, nearest railway the NASA Railroad 0.69 km, nearest highway Kennedy Parkway North 0.85 km. Launch sites sit close to open water for downrange safety and within a kilometre of rail and road links, since boosters arrive overland, yet well clear of population centres — a siting constraint any competing provider must replicate.

SpaceX launch sites and landing outcomes — 56 launch records, 24 successful landings

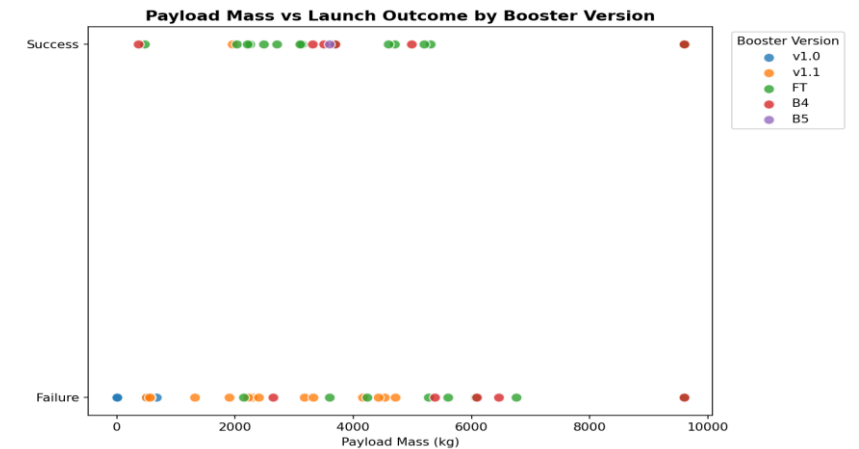
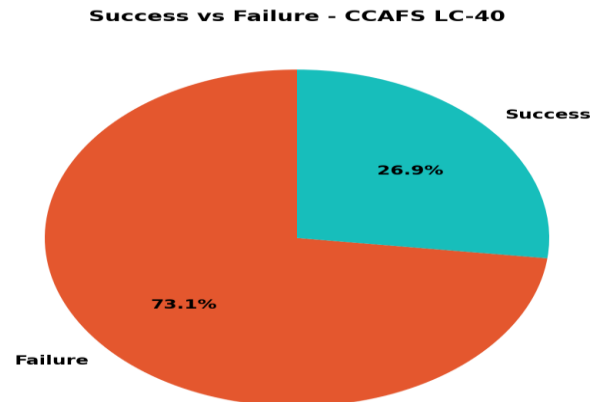
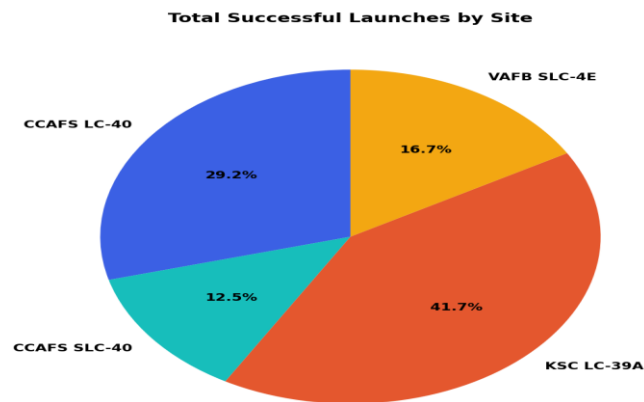


Proximity analysis — distances from KSC LC-39A



Plotly Dash App

Dropdown to filter by launch site drives a live-updating success-rate pie chart (left: share of successful landings per site; center: success/failure split at the top-volume site, CCAFS LC-40) and a RangeSlider (0–10,000 kg) narrows the payload window on the scatter plot (right), which recolors by booster version (v1.0 / v1.1 / FT / B4 / B5).



Insight: newer booster generations (FT, B4, B5) succeed reliably across a much wider payload range than the early v1.0 and v1.1 boosters — reusability maturity, not payload size alone, drives landing success. KSC LC-39A contributes the largest share of all successful landings despite launching far fewer missions than CCAFS LC-40.

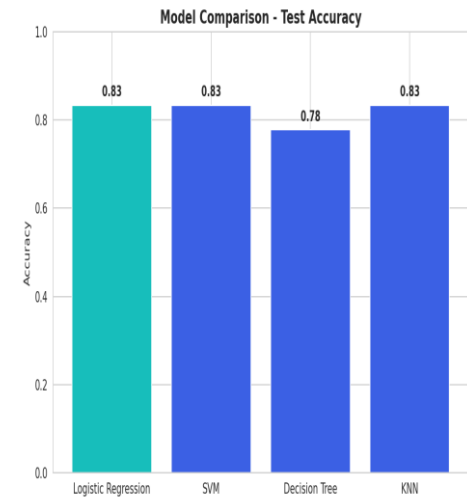
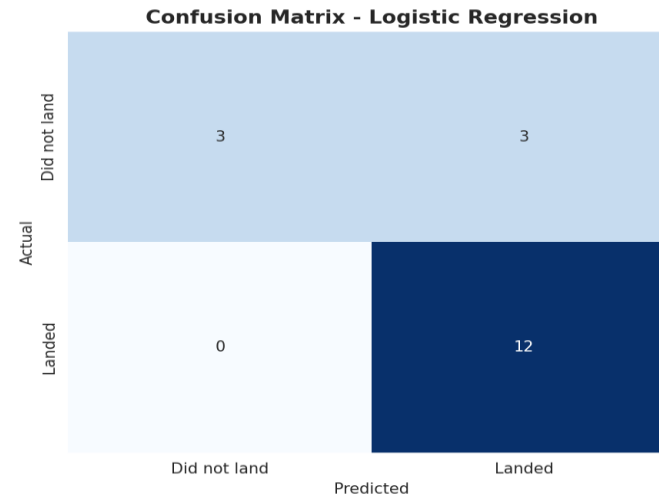
Methodology, Results & Best Model

Features: 80 one-hot-encoded columns (orbit, launch site, landing pad, booster serial) plus flight number, payload mass, flights, grid fins, reused, legs and block, standardized with StandardScaler.

Split: 80/20 train-test (72 / 18 launches).

Models, each tuned with 10-fold GridSearchCV: Logistic Regression (C, penalty, solver), Support Vector Machine (kernel, C, gamma), Decision Tree (criterion, max depth, min samples split) and K-Nearest Neighbors (n_neighbors, distance metric).

Model	CV	Test Acc.
Logistic Regression	0.821	0.833
SVM	0.834	0.833
KNN	0.834	0.833
Decision Tree	0.804	0.778



Confusion matrix (18 test launches): 12 successful landings correctly predicted, 3 unsuccessful correctly predicted, 3 unsuccessful misclassified as successful, and 0 successful landings missed — the model errs conservatively, in the safer direction for cost bidding.

Best model: Logistic Regression (C=1, L2 penalty, lbfgs solver). It ties SVM and KNN on accuracy at 83.3% but is far more interpretable — each coefficient shows a feature's effect on landing-success odds, which matters when explaining a cost bid — and it is the cheapest of the four to retrain as new launches arrive.

Conclusion & Insights

- 1 Landing success is highly learnable from public launch parameters — 83.3% test accuracy from a simple, interpretable Logistic Regression model.
- 2 Orbit type and payload mass are the strongest predictors; heavier GTO missions land far less reliably than light LEO and ISS missions.
- 3 Success rate improved dramatically over time (2013 → 2020), reflecting SpaceX's iterative process improvements rather than a static property of the hardware.
- 4 Site geography matters and is measurable: pads sit 7.43 km from the coast and within 0.85 km of rail and road, but 16.33 km from the nearest city.
- 5 Interactive tools — the Folium map and the Plotly Dash dashboard — let stakeholders verify these patterns themselves rather than trusting static charts.
- 6 Recommendation: a competing bidder can use this model as a first-pass cost-risk estimator, refining it with more recent launches and continuous retraining.

Full code, notebooks, and dashboard app: <https://github.com/razerblade09/spacex-falcon9-capstone>